

1. On the previous homework you prove Sutherland 5.1. Use that result to prove the following.
Let X be a metric space and fix $x_0 \in X$. Show that function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = d(x, x_0)$ is continuous.
2. Sutherland 5.5
3. Sutherland 5.6
4. Sutherland 5.8
5. Sutherland 5.10
6. Sutherland 5.13
7. Let X be a set endowed with the discrete metric d_{disc} . Show that a sequence (x_n) in X converges to a limit x if and only if there is $N \in \mathbb{N}$ such that if $n \geq N$ then $x_n = x$. That is: the convergent sequences are exactly the eventually constant sequences.
8. Let (X, d_X) and (Y, d_Y) be metric spaces. Show that a sequence (x_n, y_n) in $X \times Y$ converges to a limit (x, y) if and only if $x_n \rightarrow x$ and $y_n \rightarrow y$. Using the notation at the bottom of page 43, prove the result is true when $d_{X \times Y}$ is either of d_1 or d_2 . It's easy to see that it's also true for d_∞ as well, but you need not show this.
9. Suppose (x_n) is a sequence in a metric space (X, d) . Show that if $x \in X$ and if there is an $\epsilon > 0$ such that $d(x, x_n) \geq \epsilon$ for all n then $x_n \not\rightarrow x$.
10. Modify the proof from class for real valued functions to show the following.
Suppose (X, d_X) and (Y, d_Y) are metric spaces, that $f : X \rightarrow Y$ and that $a \in X$. Then f is continuous at a if and only if for every sequence (x_n) in X converging to a , $f(x_n) \rightarrow f(a)$ in Y .
11. AI Use. Please describe specifically how you used AI to help complete this assignment.